

Predictive Maintenance for Electric Vehicle Charging Station

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Abstract

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1 Problem description

1.1 Optimization model

1.2 Problem description

We consider a maintenance planning problem for a fleet of assets over a finite planning horizon. Let I denote the set of assets, F the set of failure types, and D the set of discrete time periods (days) in the horizon. For each asset $i \in I$ and failure type $f \in F$, a prognostic model provides a predicted remaining useful life RUL_{if} , interpreted as the predicted time-to-failure (in days) if no maintenance is performed.

For each relevant asset–failure pair (i, f) , the decision problem selects a maintenance execution day $d \in D$. Executing maintenance may require a setup activity (e.g., mobilization, access, tooling, or safety preparation) that incurs a fixed setup cost c_i^{setup} . We model setup as a binary decision y_{ifd} that indicates that a setup associated with failure type f on asset i is initiated on day d . Setup is assumed to occupy the asset for τ_i^s consecutive days. The maintenance execution itself is represented by x_{ifd} and occupies the asset for τ_{if} consecutive days.

Depending on the selected execution time relative to the predicted failure time, each maintenance decision is classified into exactly one of three mutually exclusive modes: *regular predictive maintenance* ($e_{if} = 1$), *urgent predictive maintenance* ($u_{if} = 1$), or *reactive maintenance* ($l_{if} = 1$), with corresponding costs c_{if}^{pr} , c_{if}^{u-pr} , and c_{if}^{re} . In addition, the model penalizes deviations from the preferred timing through nonnegative penalty variables: φ_{if} (lateness/downtime beyond RUL_{if}) weighted by α_{if} , and ψ_{if} (earliness/unused RUL before $RUL_{if} - LT_i$) weighted by β_{if} . The objective is to minimize the total cost over the planning horizon, combining setup costs, mode-dependent maintenance costs, and timing-penalty costs.

To restrict attention to failures expected within the planning horizon, we define a binary parameter F_{if} that indicates whether failure type f on asset i is predicted to occur during the horizon (i.e., $F_{if} = 1$ if $RUL_{if} \leq |D|$, and $F_{if} = 0$ otherwise). When $F_{if} = 0$, no maintenance is scheduled for (i, f) within the horizon. The complete notation is summarized below.

Sets

- I Set of assets (index i).
- F Set of failure types (index f).
- D Set of days in the planning horizon (index d).

Parameters

Maintenance and timing

- RUL_{if} Predicted remaining useful life (days) of asset i for failure type f .
- LT_i Planning lead time (days) for asset i ; maintenance earlier than $RUL_{if} - LT_i$ is penalized.
- F_{if} Indicator equal to 1 if (i, f) is within the planning horizon (e.g., $RUL_{if} \leq |D|$), and 0 otherwise.
- τ_i^s Setup duration (days) for asset i .

τ_{if}	Maintenance execution duration (days) for asset i and failure type f .
R	Daily capacity (maximum number of concurrent maintenance executions across the fleet per day).
M	Sufficiently large constant used in big- M constraints.

Costs

c_i^{setup}	Setup cost for initiating maintenance on asset i .
c_{if}^{pr}	Cost of regular predictive maintenance for (i, f) .
c_{if}^{u-pr}	Cost of urgent predictive maintenance for (i, f) .
c_{if}^{re}	Cost of reactive maintenance for (i, f) .
α_{if}	Weight for lateness/downtime penalty φ_{if} for (i, f) .
β_{if}	Weight for earliness/unused-RUL penalty ψ_{if} for (i, f) .

Decision variables

x_{ifd}	1 if maintenance for (i, f) is executed (starts) on day d , and 0 otherwise.
y_{ifd}	1 if the setup associated with (i, f) is initiated (starts) on day d , and 0 otherwise.
e_{if}	1 if (i, f) is classified as regular predictive maintenance, and 0 otherwise.
u_{if}	1 if (i, f) is classified as urgent predictive maintenance, and 0 otherwise.
l_{if}	1 if (i, f) is classified as reactive maintenance, and 0 otherwise.
φ_{if}	Nonnegative lateness/downtime amount for (i, f) (beyond RUL_{if}).
ψ_{if}	Nonnegative earliness/unused-RUL amount for (i, f) (before $RUL_{if} - LT_i$).

Thus the objective of the problem is as follows:

$$\begin{aligned}
 \text{Min } Z = & \sum_{i \in I} \sum_{f \in F} \sum_{d \in D} c_i^{setup} y_{ifd} + \sum_{i \in I} \sum_{f \in F} \alpha_{if} \varphi_{if} + \sum_{i \in I} \sum_{f \in F} \beta_{if} \psi_{if} \\
 & + \sum_{i \in I} \sum_{f \in F} c_{if}^{pr} e_{if} + \sum_{i \in I} \sum_{f \in F} c_{if}^{u-pr} u_{if} + \sum_{i \in I} \sum_{f \in F} c_{if}^{re} l_{if}.
 \end{aligned} \tag{1}$$

Objective (1) minimizes the total maintenance planning cost. It includes the setup cost c_i^{setup} whenever a setup associated with asset i is initiated on day d (via y_{ifd}), adds timing-penalty costs $\alpha_{if}\varphi_{if}$ and $\beta_{if}\psi_{if}$, and includes the mode-dependent maintenance costs: c_{if}^{pr} for regular predictive maintenance (via e_{if}), c_{if}^{u-pr} for urgent predictive maintenance (via u_{if}), and c_{if}^{re} for reactive maintenance (via l_{if}).

$$\sum_{d \in D} x_{ifd} = F_{if} \quad \forall i \in I, f \in F \quad (2)$$

$$x_{ifd} \leq y_{if, d-\tau_i^s} + \sum_{\bar{f} \neq f} x_{i\bar{f}, d-\tau_{i\bar{f}}} \quad \forall i \in I, d \in D, f \in F \quad (3)$$

$$\sum_{f \in F} \sum_{\bar{d} \in D: d-\tau_{if}+1 \leq \bar{d} \leq d} x_{if\bar{d}} + \sum_{f \in F} \sum_{\bar{d} \in D: d-\tau_i^s+1 \leq \bar{d} \leq d} y_{if\bar{d}} \leq 1 \quad \forall i \in I, d \in D \quad (4)$$

$$\sum_{i \in I} \sum_{f \in F} \sum_{\bar{d} \in D: d-\tau_{if}+1 \leq \bar{d} \leq d} x_{if\bar{d}} + \sum_{i \in I} \sum_{f \in F} \sum_{\bar{d} \in D: d-\tau_i^s+1 \leq \bar{d} \leq d} y_{if\bar{d}} \leq R \quad \forall d \in D \quad (5)$$

$$\sum_{d \in D} t_d x_{ifd} - RUL_{if} \leq \varphi_{if} \quad \forall i \in I, f \in F \mid F_{if} = 1 \quad (6)$$

$$(RUL_{if} - LT_i) - \sum_{d \in D} t_d x_{ifd} \leq \psi_{if} \quad \forall i \in I, f \in F \mid F_{if} = 1 \quad (7)$$

$$\varphi_{if} \leq M l_{if} \quad \forall i \in I, f \in F \mid F_{if} = 1 \quad (8)$$

$$\psi_{if} \leq M e_{if} \quad \forall i \in I, f \in F \mid F_{if} = 1 \quad (9)$$

$$l_{if} + e_{if} + u_{if} = F_{if} \quad \forall i \in I, f \in F \quad (10)$$

$$x_{ifd}, y_{ifd}, l_{if}, e_{if}, u_{if} \in \{0, 1\} \quad \forall i \in I, f \in F, d \in D \quad (11)$$

$$\varphi_{if} \geq 0, \psi_{if} \geq 0 \quad \forall i \in I, f \in F \quad (12)$$

Constraint (2) enforces the maintenance scheduling requirement for each asset–failure pair (i, f) . If $F_{if} = 1$, exactly one execution day d must be selected such that $x_{ifd} = 1$. If $F_{if} = 0$, then no maintenance is scheduled within the horizon and $x_{ifd} = 0$ for all $d \in D$.

Constraint (3) links maintenance execution to setup or continuation. It allows maintenance of failure type f on asset i to start on day d only if (i) a setup was initiated sufficiently early (at day $d - \tau_i^s$), or (ii) the asset is already engaged due to a recently started maintenance execution on the same asset for another failure type. The term involving $\tau_{i\bar{f}}$ captures the temporal relationship used to model continuation across failure types.

Constraint (4) enforces that, for each asset i and day d , at most one activity is ongoing on that asset. The first term counts ongoing maintenance executions (over their execution durations τ_{if}) and the second term counts ongoing setup activities (over their setup duration τ_i^s). Their sum is constrained to be at most one, preventing overlap between setup and maintenance and preventing parallel processing of multiple failures on the same asset.

Constraint (5) enforces the fleet-level daily capacity limit. For each day d , it counts the number of maintenance executions that are active on that day across all assets (accounting for their durations τ_{if}) and restricts this number to be at most R .

Constraints (6) and (7) define the timing-penalty variables. Constraint (6) upper-bounds lateness/downtime beyond the predicted failure time RUL_{if} by φ_{if} . Constraint (7) upper-bounds earliness (maintenance scheduled before $RUL_{if} - LT_i$) by ψ_{if} . These

penalty amounts are weighted in the objective by α_{if} and β_{if} , respectively. Nonnegativity is enforced by constraint (12).

Constraints (8) and (9) implement mode-dependent logical restrictions via big- M formulations. Constraint (8) allows positive lateness φ_{if} only when reactive maintenance is selected ($l_{if} = 1$). Constraint (9) allows positive earliness ψ_{if} only when regular predictive maintenance is selected ($e_{if} = 1$).

Constraint (10) enforces exclusive mode selection for each (i, f) within the horizon by requiring exactly one of the mode indicators to be active when $F_{if} = 1$ (and all to be zero when $F_{if} = 0$). Thus, each maintained failure is classified as regular predictive ($e_{if} = 1$), urgent predictive ($u_{if} = 1$), or reactive ($l_{if} = 1$).

Constraint (11) defines the binary domains of the scheduling, setup, and mode-selection variables. Constraint (12) enforces nonnegativity of the continuous timing-penalty variables.